of the lactate so produced to be converted to glycogen and the oxygen consumption rate to return to its resting level, a phenomenon known as oxygen debt.

3.8. DIABETES MELLITUS

Diabetes mellitus is the most common metabolic disease, caused by the lack of insulin activity (not secreted in sufficient amounts or does not efficiently stimulate its target cells) and results in chronic excretion of large volume of urine containing glucose.

The β cells of the pancreas produces a hormone insulin, a 51 amino acid protein in response to high blood glucose levels. Insulin acts mainly on muscle, liver, and adipose tissue cells to stimulate the synthesis of glycogen, fats, and proteins while inhibiting the breakdown of these metabolic fuels.

Insulin functions by binding to the protein receptors on the outer cell surfaces and facilitate the entry of glucose into the cells. Insulin stimulates the uptake and utilization of glucose by most cells and also elevates the rate of glycolysis, glycogen synthesis and fatty acid synthesis. Thus, it lowers the blood glucose level and together with glucagon, insulin acts to maintain the proper level of blood glucose.

The lack of insulin disrupts the regulation and balance of many metabolic pathways. As a consequence, blood glucose levels become so elevated. Yet, despite these high blood glucose levels, cells "starve" since insulin-stimulated glucose entry into cells is impaired.

3.8.1. Types

There are two major forms of diabetes mellitus:

1. **Insulin-dependent, Type 1,** or **juvenile-onset diabetes mellitus**, which most often strikes suddenly in childhood. It is caused by deficiency of pancreatic β cells or inadequate insulin secretion of insulin by β cells.
2. **Noninsulin-dependent, Type 2,** or **maturity-onset diabetes mellitus**, which usually develops rather gradually after the age of 40, mostly in overweight individuals. It is characterized by insulin resistance as well as impaired insulin secretion, where some of the insulin receptors in the cell membrane fail to recognize insulin.

3.8.2. Symptoms

The symptoms of diabetes mellitus include,

- Decreased permeability of the cell membrane for glucose resulting in the accumulation of glucose in the blood. This condition is known as hyperglycemia. Glucose concentration increases as high as 500 mg/100 ml of blood.
- **Polyuria**: This means excretion of increased quantity of urine. This is to excrete the additional quantity of glucose in urine (glucosuria).
- **Polydypsia**: The excessive thirst which leads to increased consumption of water.This condition is known as Polydypsia. This is to replace the volume of water excreted due to polyuria.

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